

Automated Location Classification Of Mining Trucks From GPS Data

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Abstract

Surface mining applications are highly dynamic environments; conditions one day may not resemble those the next. Not only are these environments subject to changes in weather and other factors, but the locations personnel and equipment operate continuously move. Conditions will also vary depending on where equipment and personnel are in the operation. Permanent locations, such as shop and crusher areas, and dynamic locations, such as material loading and dumping areas, will have different lighting, traffic, floor conditions and etc.; personnel need to be aware of these differing conditions in order to adjust behavior. While modern large-scale mining equipment is generally outfitted with GPS devices to track machines and support the optimization of operations in the mine, GPS location tracking has also become nearly ubiquitous with the increasing distribution of smart devices. The continued development and implementation of improved data processing and transmission technology increases the potential of equipment to provide more meaningful information to personnel. This research explores a methodology to analyze GPS data with a combination of clustering techniques to identify and differentiate different types of locations in the mining operation that could be applied in an appropriate tool to increase the environmental awareness of personnel.

Keywords

Clustering, GPS, Location Identification, Mining

1. Introduction

Haul Trucks represent a critical component of the material handling operation at surface mines. Rigid frame off-highway trucks are the primary mechanism of material movement from the loading tool to appropriate dumpsite and can represent up to 50% of the operating cost of a mine [1, 2]. In addition to their production and cost importance, hundreds of accidents occur annually involving mining equipment, including haul trucks [3]; in the United States alone, machinery has accounted for at least 40% of the fatalities reported by the US Mining Health and Safety Administration (MSHA) annually since 2014. It is worth noting, however, that mine injury and fatality rates as reported by the Mine Safety and Health administration have continued to decline since the late 1980s [4]. Technology solutions provide an avenue to increase the awareness of person in a mine; the Australian Center of Robotics, for example has been researching methods to extract relevant information from collision awareness systems to provide feedback to mobile equipment operators in regards to high risk traffic areas in mines [3].

The study described below will lay out a clustering methodology using GPS data to extract location information that could then be applied to provide contextual information to personnel regarding the nature of activity around them. Section 2 will summarize a review of literature and research conducted to extract activity classification information from GPS data. Section 3 will describe the proposed methodology and several of the key algorithms and techniques used in the analysis. Section 4 will provide an overview of a case study based on the GPS data collected from a North American surface hard rock mine. Sections 5 and 6 will offer a discussion and conclusion based on the results of a proposed clustering approach on adding locational context within the mine based on the analysis of GPS data.

2. Literature Review

As smart devices have increased in prevalence, so too has the research regarding the use of the data collected. Considerable focus has been spent on analyzing the context of GPS data associated with activity classification of people with data collected from smart phones. For example Gong et al. [5] utilized a two-step algorithm combining Density-Based Spatial Clustering of Applications with Noise (DBSCAN) and support vector machines to identify activity stops in urban areas; this study focused on understanding if a stop in a GPS string represents the terminus of travel, or a temporary stop along the way. Another key area of focus has been analyzing GPS traces to compare against road networks to validate road location, or generate new paths through analyses such as Principal Curve Analysis [6].

In the mining space, similar research has been more limited. The Australian Center for Field Robotics has been particularly active investigating various techniques for feature extraction from GPS data. One example of GPS analysis in the mining environment was conducted by Mao et al. [7] to track and predict vehicle positions. Much of the research found in the mining industry has been related to the extraction and updating of mine road network maps [8]. Of particular interest to this research is associated with the identification of significant places from GPS data collected from a collision awareness system installed at an Australian mine site [3]. In this research truck latitude & longitude information is separated into two distinct sets using a speed threshold, creating low-speed & high-speed locations. Activities in these locations can be seen in Figure 1(a) below. Kernel Density Estimate (KDE) contours were then overlaid atop the latitude and longitude pairs to develop a contour map of the densest areas of low-speed points; the higher the density of points, the more likely that area is significant.

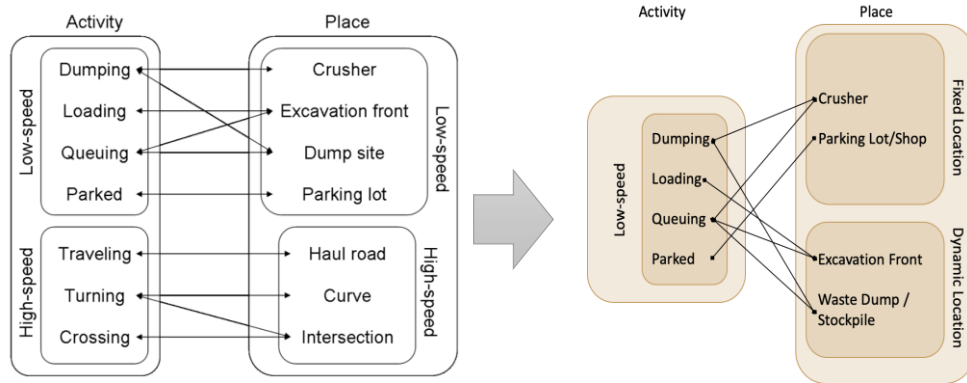


Figure 1: (a)Agamennoni Location Classification[3] & (b)Proposed Low-speed Location Classifications

A key limitation of the methodology taken by Agamennoni et al was in regard to the level of granularity taken by classifying locations, limiting the research to “significant” low-speed areas [3]. As a tool to enhance environmental awareness of personnel, this high-level classification has challenges. Loading and shop areas, for example, while both low speed areas, have different conditions for personnel to consider; shop/ready line areas are a fixed infrastructural component of the mine that is likely to have personnel on foot and significant lighting, while the loading area (excavation front) is a constantly changing area in the mine pit with generally few personnel on foot.

3. Methodology

This study intends to address the high-level limitation of the approach noted above. Similar to Agamennoni et al., this study will separate the GPS data into two groups, high speed and low speed, by applying a 5 kph speed filter. Clusters of applications below the 5 kph threshold represent key locations [3]. This study will differ first by applying a combination of K-Means and DBSCAN clustering algorithms to identify key areas in the mine. These different areas will be examined to identify Fixed and Dynamic locations, as described in Figure 1(b), intending to exploit the different spatial characteristics of each type of location; Fixed locations should be made up of a tight cluster of points, while dynamic areas should be represented by a broader distribution of points.

3.1 Data Requirements

Similar to many of the GPS feature extraction techniques examined, GPS Data requirements for this model are kept fairly simple. As is illustrated in Table 1 Truck ID, Latitude, Longitude, and Speed are the parameters required for

the model. These parameters are common in GPS devices, enabling a greater selection of hardware capable of collecting the data.

Table 1: Data Overview

Record	TruckID	Latitude	Longitude	Speed (kph)
1	1	44.076511	-103.208984	20
2	1	44.075581	-103.209373	4
3	2	44.074394	-103.20740	3
4	2	44.073654	-103.204689	6

3.2 K-Means Algorithm

The K-Means algorithm represents one of the most common clustering algorithms available. This unsupervised clustering algorithm divides a dataset into a predetermined number of clusters, k . In summary, the algorithm randomly divides the data into k clusters, then calculates the distance between each point and the centroid of each cluster. Each point is then reassigned to a cluster based on the nearest centroid. The centroid is then recalculated and the distances between points and centroids calculated again. These steps are repeated until cluster assignments stabilize [9].

One potential drawback of K-means algorithm for some analyses is the requirement to predetermine the value of k . This study explores the automation of cluster count selection through the application of the Silhouette method, which is designed to measure the validity of a clustering analysis. This method can be used to automate the selection of the number of clusters by running a series of K-Means analyses, calculating the Silhouette score for each analysis and selecting the k -value with the highest score. In summary, a Silhouette analysis calculates the normalized difference between a point's mean distance ($a_{p,i}$) from every other point in its cluster, p , and the minimum mean distance ($b_{q,i}$) between that point all other clusters, q , and normalized with the total number of points in the analysis (N) as described in Formula 1 [10].

$$Silhouette = \frac{1}{N} \sum_{i=0}^N \frac{(b_{q,i} - a_{p,i})}{\max\{a_{p,i}, b_{q,i}\}} \quad (1)$$

3.3 DBSCAN Algorithm

The Density-Based Spatial Clustering of Applications with Noise (DBSCAN) clustering algorithm was developed by Ester et al. [11] as a method for efficiently grouping spatial data into meaningful groups and identifying noise. Unlike many partitioning algorithms, such as K-Means, DBSCAN does not require upfront definition of the number of clusters. Instead the algorithm groups points together based on the density of points in a particular area. Those points that are in low density areas are classified as noise. Key inputs to this algorithm are epsilon, representing the maximum distance between points, and definition of the minimum number of points in a cluster.

A key element of the density calculation in DBSCAN is the calculation of distance between points. The DBSCAN algorithm is agnostic to the distance calculation, allowing the analyst to use any calculation that best fits the project [11]. In regards to this methodology, global generalization requires a distance calculation compatible with latitude (ϕ) and longitude (ψ) coordinates. The haversine formula, a common navigational method for calculating great circle distances, expressed below, was used to calculate distance (d) between two points utilizing the radius of the Earth (r) [12].

$$d = 2r \sin^{-1} \left(\sqrt{\left(\sin \frac{\phi_2 - \phi_1}{2} \right)^2 + \cos \phi_1 \times \cos \phi_2 \times \left(\sin \frac{\psi_2 - \psi_1}{2} \right)^2} \right) \quad (2)$$

One drawback of the DBSCAN algorithm lies in its computation time. Since this algorithm is based on clustering data based off the distance of each point to the next, computation time can quickly increase as the data set increases. Several different performance optimized variants of DBSCAN have been developed, such as IDBSCAN[13]. The general DBSCAN algorithm will be used in this analysis, however, due to its availability in analytics toolsets and because computational performance is not a primary target of this study.

3.4 Area Definition

Area boundaries, for use in the calculation of cluster geometries, were determined using the Quickhull variant of the Convex hull method [14]. In brief, for a set of 2 dimensional points, x & y , the points with maximum and minimum x points are found, added to the list of boundary points and a line is made between these points. The dataset is then

divided in two based on the bisecting line. For one of the data sets, the point furthest from the line is calculated and added to the list of boundary points. Lines are then calculated from the initial maximum and minimum x points to the new boundary point, and the dataset again divided and points farthest from the new lines drawn and farthest points determined. This process is repeated until no new points can be added. These boundaries, once calculated, can be used to provide an array of boundary points that can be used to calculate various geometric statistics about the clusters for comparison.

3.5 Consolidated Algorithm

This study proposes to combine K-Means and DBSCAN algorithms in order to identify significant locations in a mine. K-Means, with automated k selection with Silhouettes, will be applied individually for each truck in the fleet. The centroids of each K-Means analysis will be combined and processed with DBSCAN to determine final clusters; to improve significance of certain centroids, each K-Means centroid will be relatively weighted based on how many points from the total dataset are assigned to that cluster. Once completed, the geometric statistics of each cluster will be calculated for comparison to identify the nature of each significant location. This algorithm is summarized in Figure 2.

1. Filter GPS to Speeds < 5kph
2. For TruckID in Fleet:
 - A. For i in range x:
 - a. Perform K-Means Cluster where $k=i$
 - b. Calculate Silhouette
 - B. Use max(centroid)
3. Calculate Centroids of K-Means Clusters
4. Consolidate K-Means Clusters Data
5. Weight K-Means Centroids
6. Perform DBSCAN on consolidated K-Means centroids
7. Calculate Distances between Points in Each DBSCAN Cluster
8. Calculate Boundary of each Cluster using Convex Hull

Figure 2: Algorithm Pseudocode

The combination of K-Means and DBSCAN allows for enhancement of each algorithm and a spatial narrowing of the dataset. The iterative K-Means algorithm by truck reduces the data set to an array of centroids representative of where the machines generally spend low-speed time. DBSCAN then clusters these centroids into areas of key significance, using its noise classification feature to identify areas that do not have significance to the fleet as a whole.

4. Case Study

4.1 Data Used

The GPS data utilized for this analysis was collected from the maintenance management tool at a North American hard rock mine and represents roughly one day of data for 17 trucks. These GPS points were only collected when the trucks were running at a rate of 5 seconds per sample. This maintenance management tool only collects data from running machines, therefore total data quantities vary machine to machine. Since troubleshooting can require the running of equipment, there is significant data in the shop area for one truck. In total 250,136 points were available for the analysis, of which 10% was utilized after random sampling. The features utilized are limited to those shown in Table 1: latitude, longitude, vehicle speed and truck id.

4.2 Modelling

The layered model described in section 3.5 was compared against Silhouette automated K-Means and basic DBSCAN clustering. In both DBSCAN models an epsilon of 100 meters was used based on observed queue length at a shovel at the mine; this figure should be adjusted based on equipment selection and operational practices of the mine.

4.3 Results

Key results of the methods described above are reported in Figure 3, a graphical comparison of the three clustering methods. A third fixed location, “E Non Action” (East Non Action) was identified for further analysis; this location, to the east of the crusher, represented a high concentration of low-speed points in all three clustering methods that had no corresponding infrastructure or loading or dumping event. A degree of focus is placed on the fixed locations when studying this model; since the dynamic areas are transitory in nature, different numbers or sizes of clusters can be developed with different methodologies. The fixed locations, however, are unambiguous.

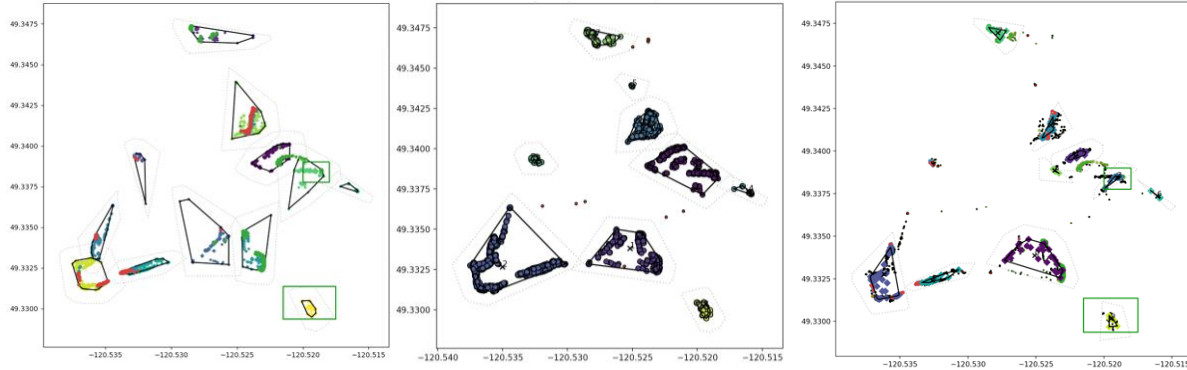


Figure 3: Cluster Method Comparison (left to right: K-Means, DBSCAN, Layered)

Evaluation of the three models was based on a comparison against a test dataset consisting of GPS tagged loading and dumping events from a separate system, as well as input regarding the location of the fixed crusher and shop locations. The machine generated events were clustered with K-Means and the labeled data combined with the input fixed location coordinates. The centroids of the model and test data clusters were then calculated. The model’s classification was compared to the classification of the closest test cluster, and summarized in Table 2. Confusion matrices were then developed to evaluate separately the success of the prediction of fixed and dynamic locations. Recall, precision and their harmonic mean, F-Measure, were selected as primary scoring measures and are shown in Table 3 [15].

Table 2: Classification Overview

Clustering Method	Total Clusters	Count Fixed Location		Count Dynamic Location	
		Correct	Incorrect	Correct	Incorrect
K-Means	12	2	0	9	1
DBSCAN	9	2	2	4	1
Layered	10	3	3	4	0

Table 3: Classification Scores

Clustering Method	Fixed Location			Dynamic Location		
	Recall	Precision	F-Measure	Recall	Precision	F-Measure
K-Means	67%	100%	80%	100%	90%	95%
DBSCAN	67%	50%	57%	67%	80%	73%
Layered	100%	50%	67%	57%	100%	73%

As expected, the basic K-Means algorithm produced larger clusters incorporating all GPS data. While efficient in processing, incorporating all low speed GPS data doesn’t distinguish between legitimate mine locations and low speed points that may arise from traffic patterns or periodic machine stops along the road. Straight DBSCAN categorized some of the low speed points as noise, but was more computationally intensive. Also, the GPS point cluster associated with the shop area included a nearby dump area correctly classified by the other two methods. The Layered methodology identified all 3 fixed locations, but was also the only algorithm to generate several false fixed locations.

The Layered methodology also produced tighter clusters than K-Means, identifying more noise points (~5.8%) that were not near the test Loading and Dumping events.

5. Conclusions

Initial results indicate Layered clustering methodology may demonstrate tighter cluster definition than either K-Means or DBSCAN algorithms alone, improving on noise reduction from K-Means and cluster differentiation from DBSCAN. This would support more focused feedback regarding location activity if incorporated in a personnel awareness system. Differentiation between dynamic and fixed locations for the proposed model requires further refinement; while being the only algorithm to successfully identify all three fixed areas, the layered approach also falsely classified several of the dynamic areas.

Regarding future research, continued evaluation of the Layered clustering method described above will include further refinement of parameters differentiating fixed from dynamic mine locations; elevation, for example, properly applied may help differentiate different mine areas. Further analysis using different GPS data will be conducted to test the generalization of the model. Deeper analysis into the nature of more specific fixed and dynamic mine locations would provide more context regarding the environment in different areas of a mine; for example, whether or not a dynamic area represents a loading front.

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